

Exploring Cluster Structures through Interactive Attribute Weighting

A Visual Analytics Interface for Clustering Evaluation

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Motivation

Problem

Clustering results are often difficult to interpret because the user does not know which attributes actually drive the grouping of data points.

- Different attributes can strengthen, weaken, or distort cluster separation.
- A fixed projection hides how sensitive the result is to feature importance.
- Users need both **quantitative feedback** and **interactive visual exploration**.

Goal

Provide a visual analytics tool where users can change attribute weights, observe MDS projections, compare clustering views, and inspect meaningful configurations.

Core Idea

- The dataset is projected through **Multidimensional Scaling**.
- Each numerical attribute receives a weight $w_k \in [0, 1]$.
- Changing weights changes pairwise distances and therefore the projection.



The user can test hypotheses interactively.

- Low weight: attribute has little influence.
- High weight: attribute strongly affects similarity.

Weighted distance

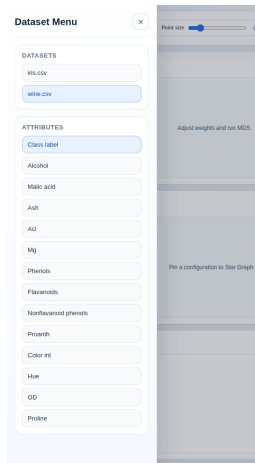
$$d_w(i, j) = \sqrt{\sum_{k=1}^p w_k (x_{ik} - x_{jk})^2}$$

Workflow

1. **Select or upload a dataset:** default datasets are *wine.csv* and *iris.csv*.
2. **Choose the label attribute:** the first attribute is selected by default, but the user can change it.
3. **Adjust feature weights:** non-categorical attributes are weighted in steps of 0.05.
4. **Observe projections:** the interface updates MDS-based clustering views.
5. **Save and compare configurations:** Silhouette values and Star charts support comparison.
6. **Inspect local behavior:** lasso selection highlights the same points across projections.

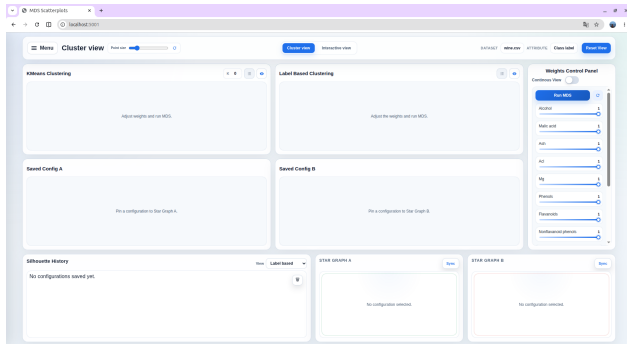
Dataset and Label Selection

- The user can decide which dataset attribute should act as the reference label.
- The selected label is excluded from some weighted features.
- This makes the analysis adaptable to different interpretation goals.



Cluster View

- Four scatter plots summarize the current and saved projections.
- *Top row*: current weighted MDS projection.
- K-Means and label-based clustering are shown side by side.
- *Lower row*: contains saved configurations selected from the history.



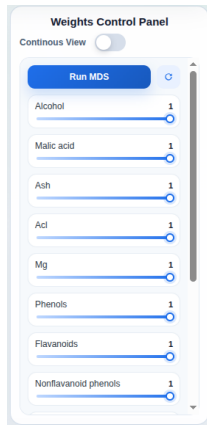
Weight Control Panel and Continuous View

Weight Control Panel

Lists the available attributes and allows the user to tune each coefficient between 0 and 1.

Continuous View

Automatically updates the projection while the user changes feature weights.

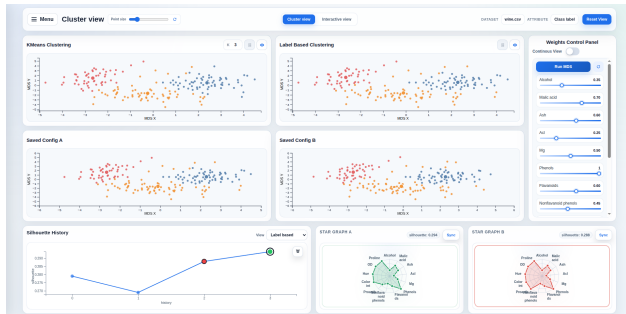


Continuous View features

- Sends each weight change to the server.
- Recomputes weighted distances and MDS coordinates in real time.
- Supports fast exploration without repeatedly pressing *Run MDS*.
- The user can still save only the meaningful configurations.

Saving and Comparing Configurations

- The **Silhouette Chart** stores saved weighting configurations.
- Each point is associated with a silhouette value.
- Star graphs show the weight distribution of selected configurations.
- **Sync** buttons restore saved settings and display their projections.



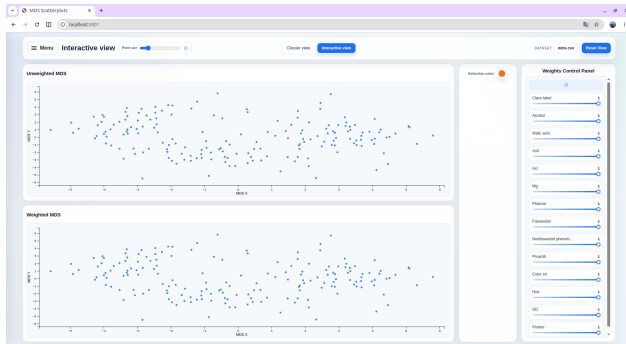
Silhouette History

- The **Silhouette value** gives a compact measure of *cluster quality*.
- Higher values suggest better separation and compactness.
- The chart helps identify promising configurations during exploration.



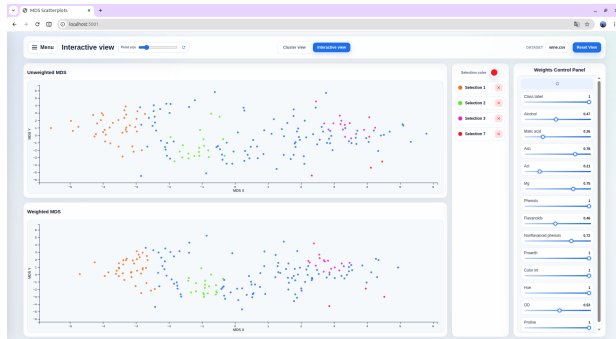
Interactive View

- The **Interactive View** compares unweighted and weighted MDS projections
- The weighted projection changes live when the user modifies coefficients
- This view focuses on *direct exploration* rather than only global cluster scores.



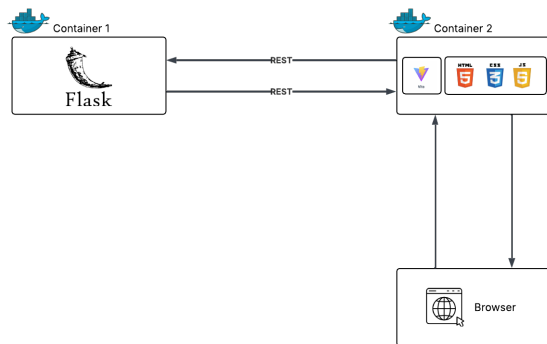
Interactive View Features

- The user selects a region of points with a lasso tool.
- The same points are highlighted in both projections.
- This reveals whether a subset is stable, separated, or ambiguous under weighting.



System Architecture

- The system is organized in two Docker containers.
- **Flask server:** preprocessing, weighted distances, MDS, clustering, silhouette.
- **Vite front-end:** interactive visual interface in the browser.
- The browser handles interaction while the server manages computation.



Insight: What the User Can Learn

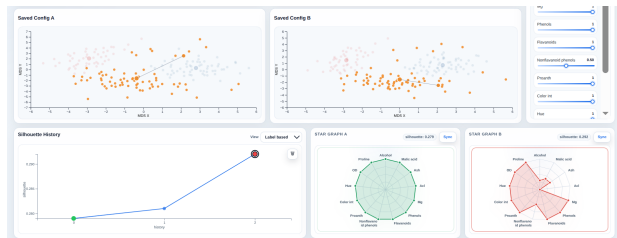
- Which attributes are influential for the similarity relationships.
- Whether clusters remain stable when weights change.
- Whether automatic K-Means clusters agree with selected labels.
- Which points behave like outliers or ambiguous cases.
- Whether a 2D MDS projection is stable or visually misleading.

Interpretation principle

Reliable interpretation comes from combining point movement, cluster separation, Silhouette values, star graphs and projection stability.

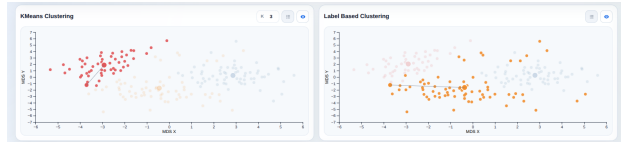
Use Case 1: Influential Attributes

- Weight tuning can improve cluster separation.
- An increased Silhouette value indicates a better configuration.
- Comparing saved configurations helps identify attributes that support the cluster structure.



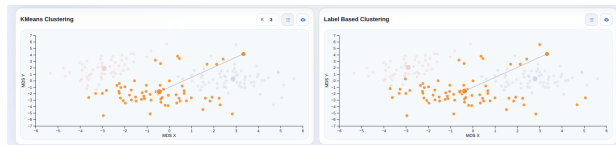
Use Case 2: Outliers

- A point may belong to one label-based group, but it can be assigned by K-Means to another cluster.
- This mismatch suggests a possible outlier or ambiguous data point.
- The user can inspect whether the weights reveal hidden relationships.



Use Case 3: Projection Instability

- A point can remain in the same cluster but appear far from its centroid in the MDS view.
- This suggests that the assignment may be stable, while the 2D projection is distorted.
- This use case highlights why visual interpretation must be combined with quantitative indicators.



Conclusions

- The application turns clustering evaluation into an interactive Visual Analytics workflow.
- Attribute weights help users understand how much each attribute contributes and which attributes are the most informative.
- MDS projections, Silhouette history, star graphs and lasso selection provide complementary evidence.
- The tool helps identify influential attributes, outliers, ambiguous points, and projection instability.

Main takeaway

The system does not only show a clustering result: it helps explain *why* the result changes when the user changes the importance of the attributes.